LECTURE 2 — TENSOR COMPONENTS

2.1 Bases and vector components

Def. A subset $B \subset V$ for a vector space $(V, \oplus, \odot)$ is called linearly independent if for any finite subset $\{e_1, e_2, \ldots, e_n\} \subseteq B$ the condition

$$\sum_{i=1}^{\dim V} \lambda^i \odot e_i = O \text{ requires } \lambda^1 = \lambda^2 = \cdots = \lambda^n = 0 \ .$$

Def. Such B is called a basis for the vector space if for any $v \in V$ there is a subset $\{e_1, \ldots, e_{n_v}\} \subset B$ such that

$$v = \sum_{i=1}^{\dim V} v^i \odot e_i$$

for (then unique) components $v^1, \ldots, v^n \in \mathbb{R}$

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Example. Vector space $(\mathbb{R} \times \mathbb{R}, \oplus, \odot)$ as before

$B = \{(1,2),(1,3)\}$ is a basis

Thus there is a unique decomposition of,

e.g., $(5,13) = 2 \odot (1,2) \oplus 3 \odot (1,3)$

components of. $(5,13)$ w.r.t. B

$B' = \{(1,0),(0,1)\}$ is (another) basis

Now $(5,13) = 5 \odot (1,0) \oplus 13 \odot (0,1)$

components of. $(5,13)$ w.r.t. B'

Anyway: $\dim \mathbb{R} \times \mathbb{R} = 2$

2.2 Dual basis

Thm. For any choice of basis

$$B = \{e_1, \ldots, e_{\dim V}\}$$

of a finite-dimensional vector space $(V, \oplus, \odot)$

there is a unique „dual“ basis

$$B^* = \{\epsilon^1, \ldots, \epsilon^{\dim V}\}$$

for the dual vector space $(V^*, \boxplus, \boxdot)$

defined by the condition

$$\epsilon^i(e_j) = \delta^i_j.$$

Remark. It is economical to link the basis of $(V^*, \boxplus, \boxdot)$ to the chosen basis of $(V, \oplus, \odot)$ in this way.

Application

The components of $v \in V$ w.r.t. a basis $e_1, \ldots, e_{\dim V}$ can be directly calculated as

$$v^i := \epsilon^i(v).$$

Proof:
$$\begin{aligned}
\epsilon^i(v) &= \epsilon^i\left(v^1 \odot e_1 \oplus \cdots \oplus v^{\dim V} \odot e_{\dim V}\right) \\
&= \epsilon^i(v^1 \odot e_1) + \cdots + \epsilon^i(v^{\dim V} \odot e_{\dim V}) \\
&= v^1 \epsilon^i(e_1) + \cdots + v^{\dim V} \epsilon^i(e_{\dim V}) \\
&= 0 + \cdots + v^i \epsilon^i(e_i) + \cdots + 0 \\
&= v^i
\end{aligned}$$

In other words:
$$v = \sum_{i=1}^{\dim V} \epsilon^i(v) \odot e_i$$

2.3 Components of tensors

Generalization of the previous application

Def. The <u>components of a tensor</u>

$$T : \underbrace{V^* \times \cdots \times V^*}_{p} \times \underbrace{V \times \cdots \times V}_{q} \to \mathbb{R}$$

are given by the real numbers

$$T^{i_1 \ldots i_p}{}_{j_1 \ldots j_q} := T(\epsilon^{i_1}, \ldots, \epsilon^{i_p}, e_{j_1}, \ldots, e_{j_q}).$$

Example. Let $T : V^* \times V \to \mathbb{R}$ be a (1,1)-tensor

$$T(\varphi, v) = T\Big(\mathop{\boxed{+}}_{i=1}^{\dim V} \varphi_i \boxdot \epsilon^i, \bigoplus_{j=1}^{\dim V} v^j \odot e_j \Big) = \sum_{i=1}^{\dim V} \sum_{j=1}^{\dim V} \varphi_i T^i{}_j v^j$$

Can calculate $T(\varphi, v)$ in terms of <u>components</u> of T, φ, v.

2.4 Changes of basis

Affects *components* of a tensor, but <u>not</u> the tensor.

Thm. Let $e_1, \ldots, e_{\dim V}$ and $\tilde{e}_1, \ldots, \tilde{e}_{\dim V}$ be bases.

Then
$$\tilde{e}_j = \bigoplus_{a=1}^{\dim V} C^a{}_j \odot e_a$$

and thus
$$\tilde{\epsilon}^{\,i} = \boxplus_{b=1}^{\dim V} (C^{-1})^i{}_b \boxdot \epsilon^b \quad .$$

Thm. Tensor components w.r.t. new bases $\tilde{e}_j$ and $\tilde{\epsilon}^{\,i}$ are

$$\tilde{T}^{i_1 i_2}{}_{j_1} = \sum_{a_1=1}^{\dim V} \sum_{a_2=1}^{\dim V} \sum_{b_1=1}^{\dim V} (C^{-1})^{i_1}{}_{a_1} (C^{-1})^{i_2}{}_{a_2} C^{b_1}{}_{j_1} T^{a_1 a_2}{}_{b_1}$$